Federated Learning for Soybean Disease Classification

Comparison of CNN and Vision Transformer Architectures

Training Results Report

March 24, 2026

1 Executive Summary

This report presents the results of federated learning experiments for soybean disease classification across seven disease classes. Four deep learning architectures—ViT-Small, EfficientNet-B0, MobileNet-V2, and ResNet-18—were trained for 50 federated rounds using the Flower framework with the FedAvg strategy.

Key Findings:

- **ViT-Small** achieves the highest test accuracy (**91.25%**) and macro F1-score (**0.872**), significantly outperforming all CNN-based models.
- **EfficientNet-B0** and **MobileNet-V2** show moderate performance (43.76% and 66.23% test accuracy respectively), while **ResNet-18** underperforms at 55.31%.
- Class imbalance remains a challenge: several classes achieve near-perfect recall across all models, while *Downey Mildew* and *Frogeye* show the weakest performance for weaker models.
- ViT-Small required the longest training time (280.500 min) but delivered substantially better generalization, indicating a favorable accuracy–cost trade-off.

2 Experimental Setup

Table 1: Hyperparameters shared across all experiments.

Parameter	Value
FL Framework	Flower
FL Strategy	FedAvg
Communication Rounds	50
Local Epochs	3
Batch Size	8
Learning Rate	3.000×10^{-5}
Device	NVIDIA RTX 3050 Laptop GPU

The dataset contains images from seven soybean disease classes: Bacterial Blight, Cercospora Leaf Blight, Downey Mildew, Frogeye, Healthy, Soybean Rust, and Target Spot.

3 Aggregate Model Comparison

Table 2 summarizes the final test-set metrics for each model.

Table 2: Final evaluation metrics on the held-out test set.

Model	Val Acc	Test Acc	Macro F1	Wtd F1	Macro Prec	Time (min)
ViT-Small	0.902	0.913	0.872	0.906	0.925	280.500
EfficientNet-B0	0.438	0.438	0.381	0.406	0.553	177.600
MobileNet-V2	0.654	0.662	0.506	0.581	0.505	131.100
ResNet-18	0.569	0.553	0.382	0.488	0.504	132.000

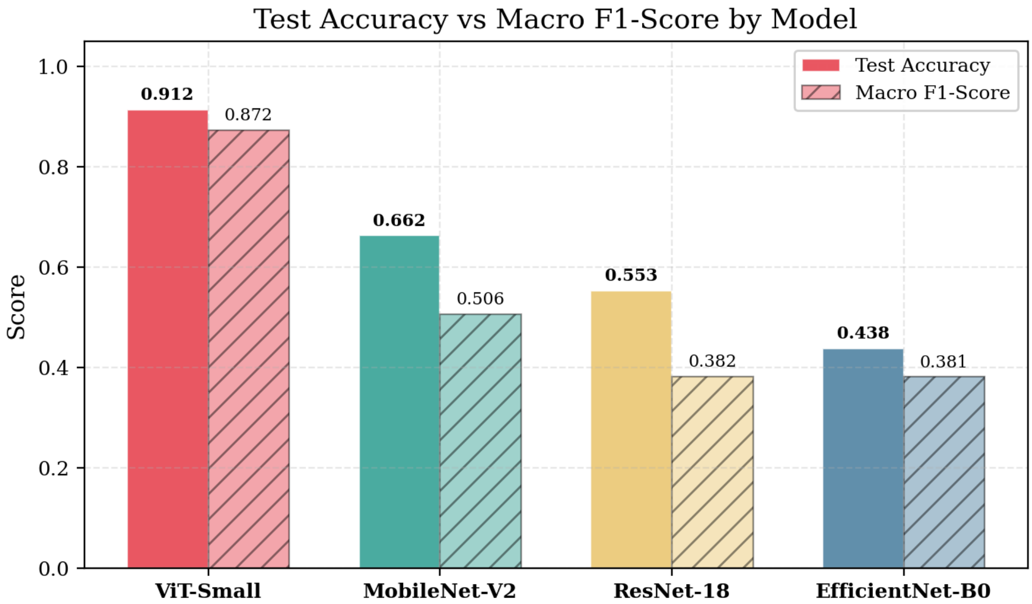

Figure 1: Test accuracy and macro F1-score grouped by model. ViT-Small dominates both metrics.

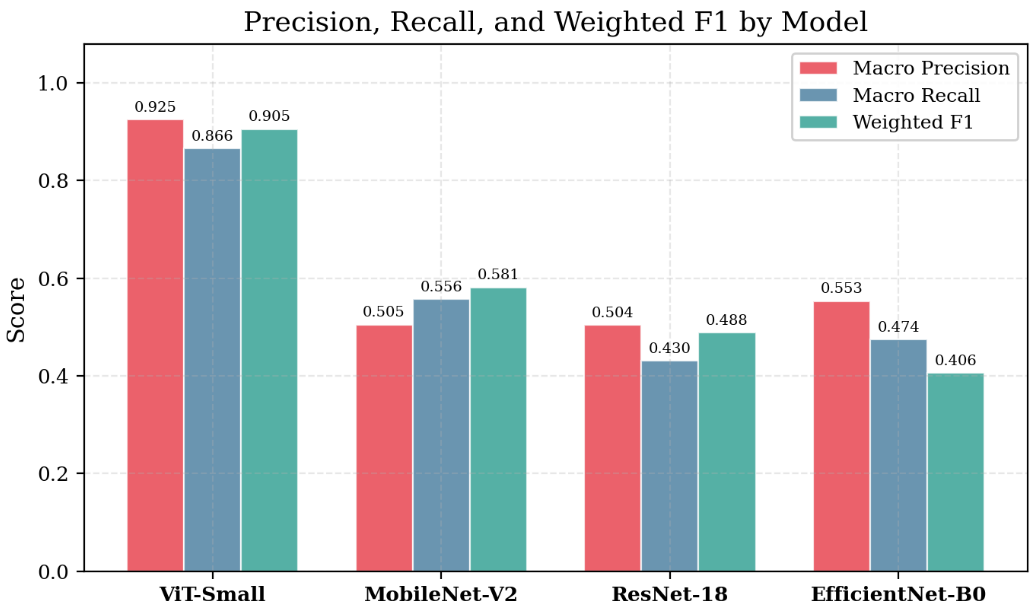

Figure 2: Macro precision, macro recall, and weighted F1 for each model.

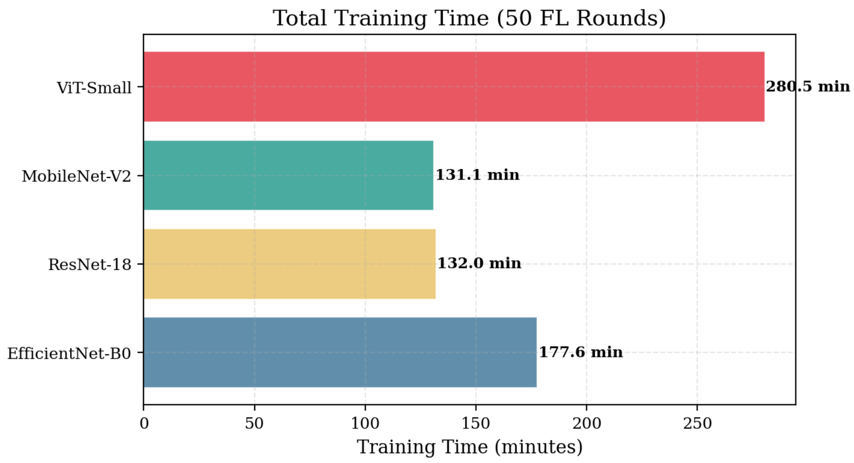

Figure 3: Total training time across 50 federated rounds.

4 Training Dynamics

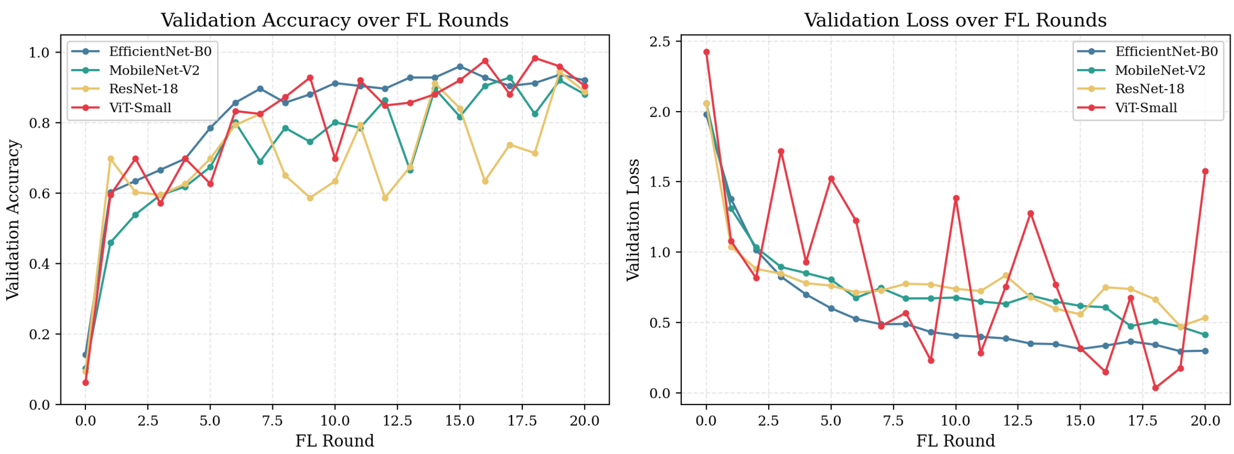

Figure 4: Validation accuracy and loss over 50 federated rounds for all models.

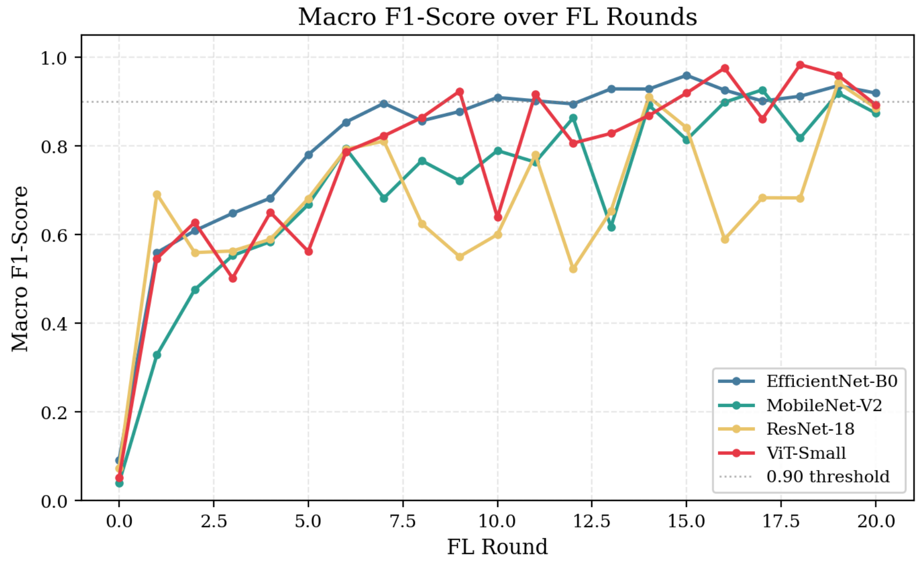

Figure 5: Macro F1 convergence over rounds. The dashed line marks the 0.90 threshold.

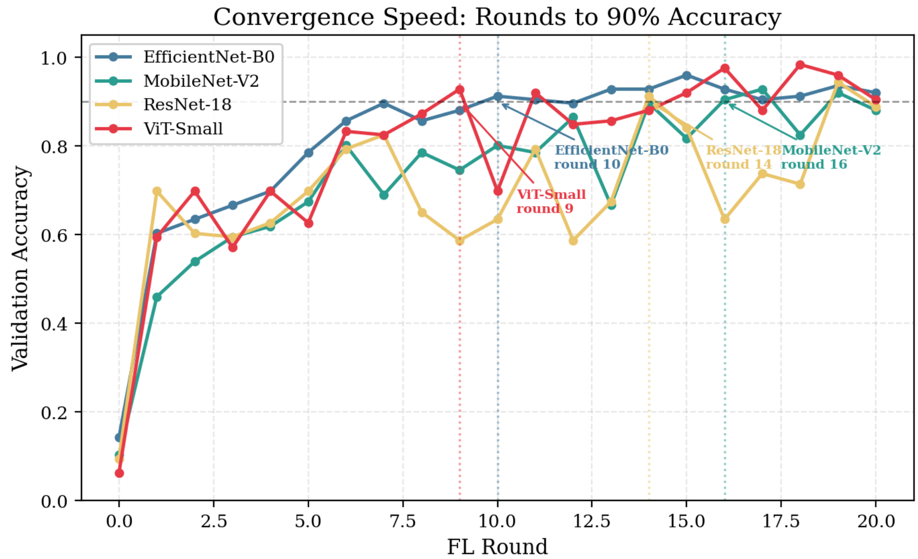

Figure 6: Convergence analysis showing the first round each model crosses 90% validation accuracy.

ViT-Small reaches 90% accuracy by round 16 and continues to improve, achieving 97.6% at its peak. EfficientNet-B0 shows a steadier climb but stalls below 50% overall test accuracy. MobileNet-V2 exhibits high variance between rounds, suggesting sensitivity to client data distributions.

5 Per-Class Analysis

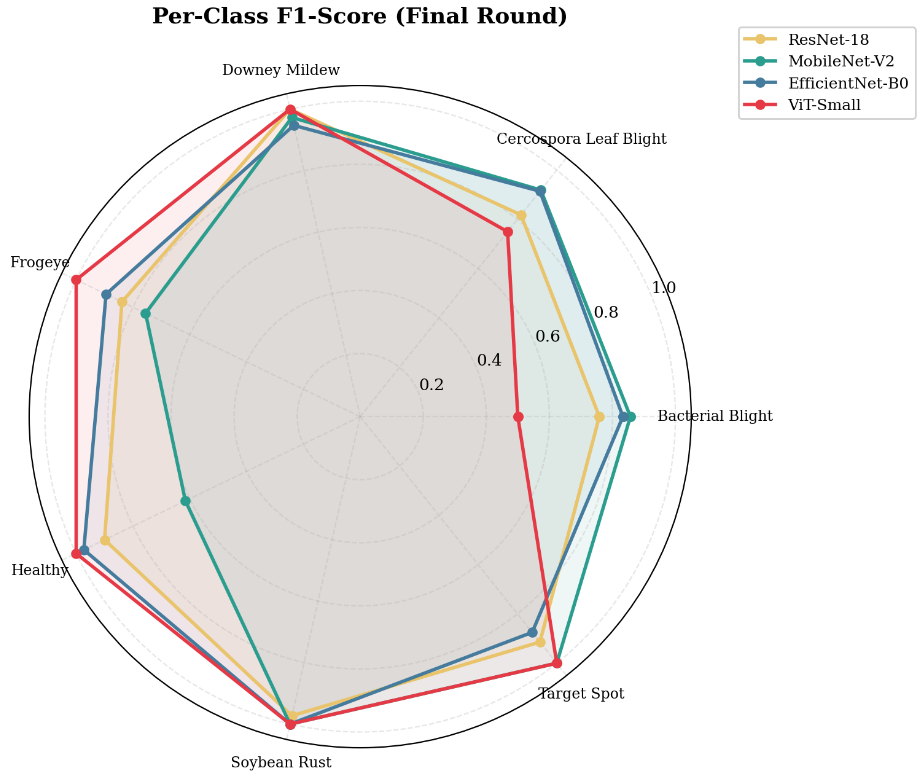

Figure 7: Radar chart of per-class F1 scores at the final round for all models.

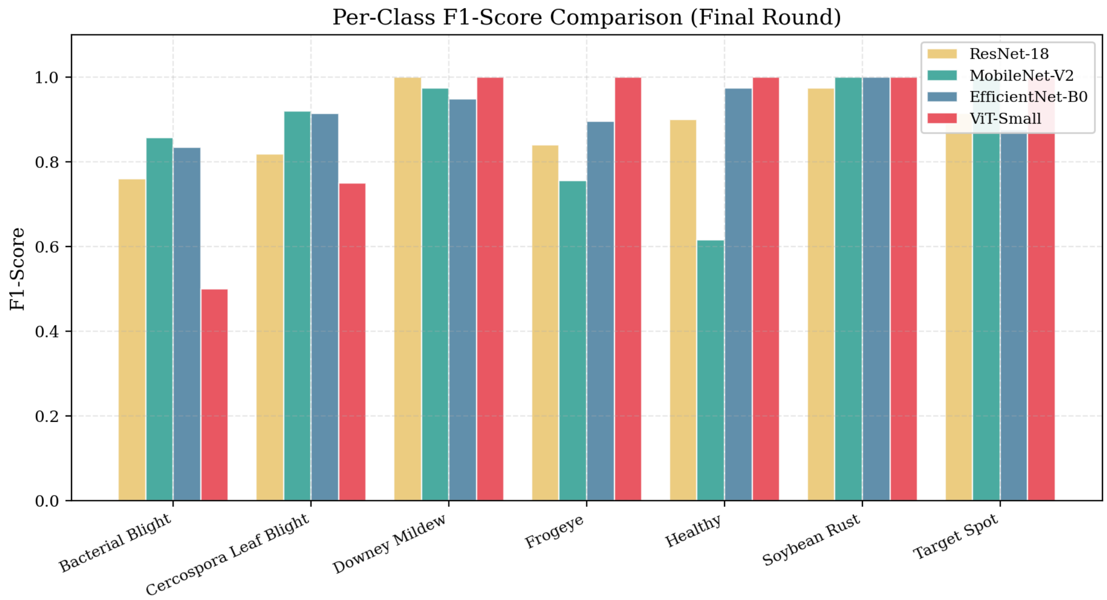

Figure 8: Grouped bar chart of per-class F1-scores at the final round.

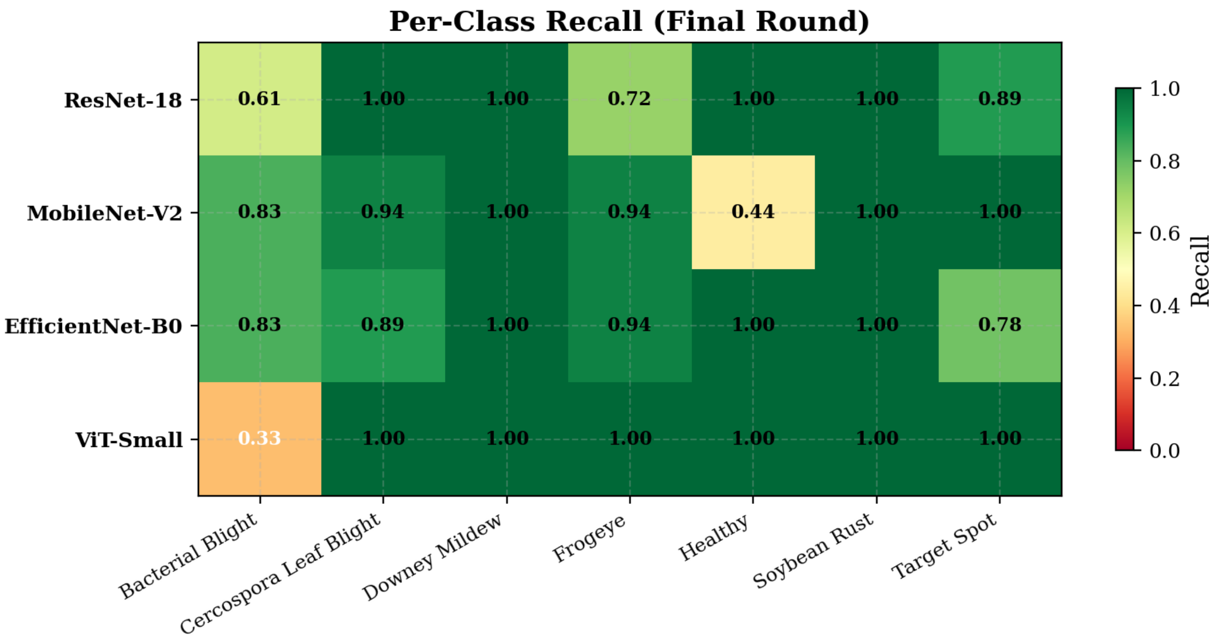

Figure 9: Per-class recall heatmap (final round). Green indicates strong recall; red indicates weak recall.

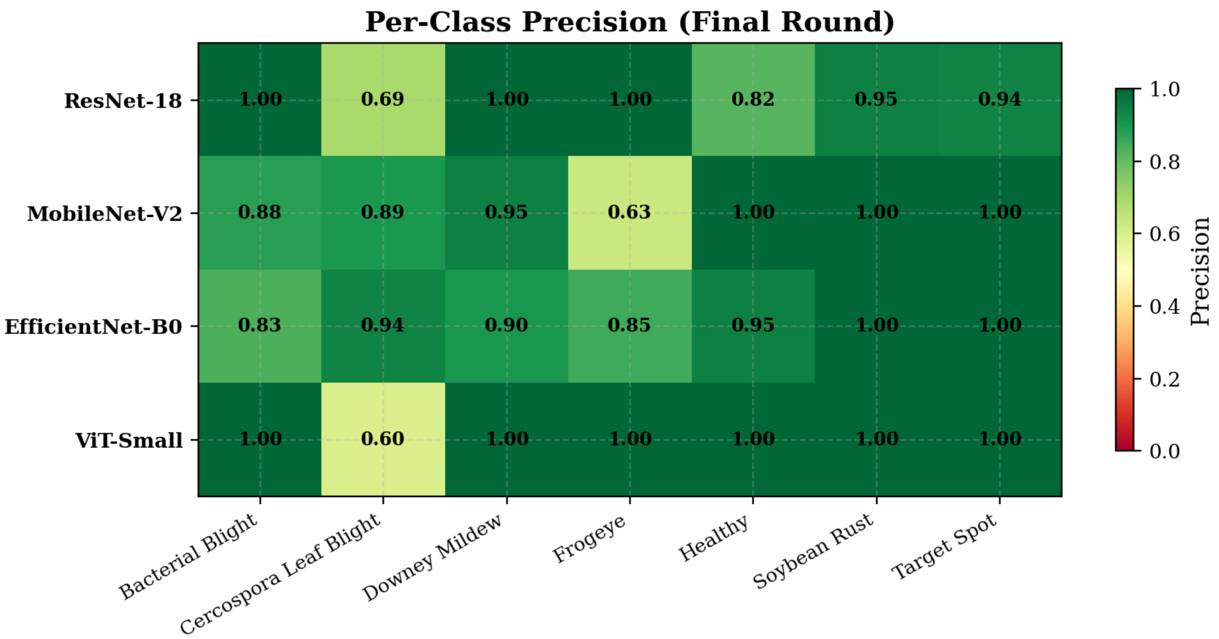

Figure 10: Per-class precision heatmap (final round).

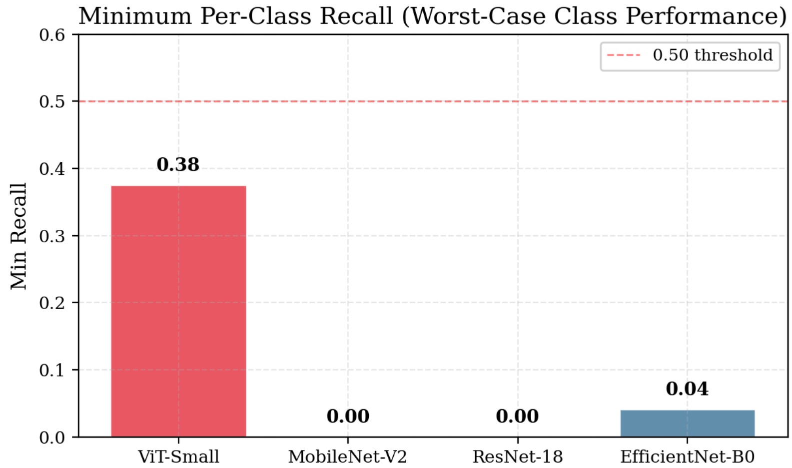

Figure 11: Minimum per-class recall across all classes. Higher is better (no class left behind).

Observations:

- **ViT-Small** achieves perfect or near-perfect F1 on most classes, with the lowest performance on *Bacterial Blight* (F1 = 0.50) and *Cercospora Leaf Blight* (F1 = 0.75).
- **ResNet-18** and **MobileNet-V2** struggle significantly with *Downey Mildew* and *Frogeye*, where recall drops to zero or near-zero for weaker rounds.
- *Healthy* and *Target Spot* classes are well-recognized across all architectures.

6 Efficiency Analysis

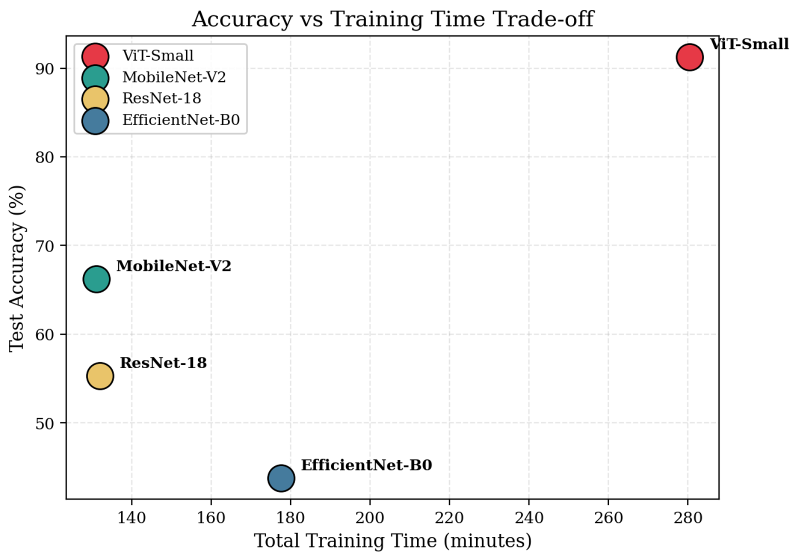

Figure 12: Accuracy vs. training time trade-off. Upper-left is ideal (high accuracy, low time).

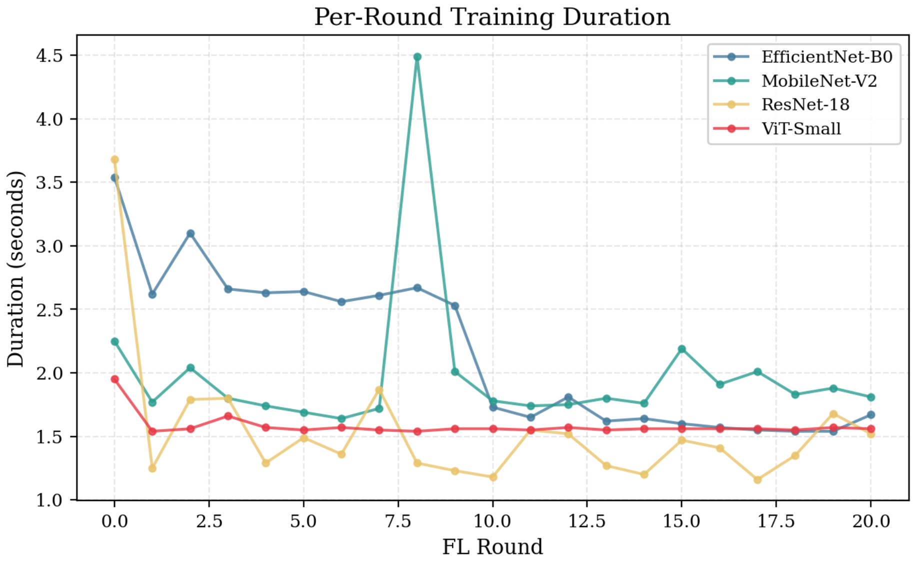

Figure 13: Per-round training duration. ViT-Small rounds are slightly longer due to higher parameter count.

Table 3: Efficiency metrics: accuracy per minute of training and parameter counts.

Model	**Time (min)**	**Test Acc**	**Acc / min**
ViT-Small	280.500	0.913	0.003
EfficientNet-B0	177.600	0.438	0.002
MobileNet-V2	131.100	0.662	0.005
ResNet-18	132.000	0.553	0.004

While MobileNet-V2 offers the best accuracy-per-minute ratio, its absolute accuracy is far below ViT-Small. For applications demanding high reliability, ViT-Small's additional training cost is justified.

7 Dashboard Summary

Figure 14: Comprehensive dashboard summarizing test accuracy, F1-score, learning curves, and per-class radar chart.

8 Conclusions and Recommendations

1. **ViT-Small is the clear winner** for this federated soybean disease classification task, achieving 91.25% test accuracy and a macro F1 of 0.872—a 25+ percentage point gap over the next-best CNN model.

2. **Class imbalance persists** as a challenge. EfficientNet-B0, ResNet-18, and MobileNet-V2 all report zero minimum per-class recall, indicating complete failure on at least one disease class. Future work should explore class-weighted loss functions or oversampling of minority classes.

3. **Convergence is fast for ViT-Small** (reaching 90% by round 16) but slow and unstable for the CNN models, suggesting that the Vision Transformer architecture is more robust to heterogeneous data distributions across federated clients.